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A multimodal machine learning approach to predict Fugl-Meyer scores and motor recovery potential in stroke rehabilitation: Toward precision-based therapies*

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Abstract

Stroke is a leading cause of long-term disability, with highly variable recovery trajectories and challenges in prediction and monitoring. Frequently used measures (e.g., National Institute of Health Stroke Scale (NIHSS) and Fugl-Meyer (FM) assessment of motor impairment) have significant limitations. As the societal burden of stroke increases, developing robust methodologies for assessing and predicting recovery is essential to optimize treatment plans and improve outcomes.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT/Microsoft Co-Pilot in order to improve readability of select parts of the manuscript. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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This paper presents our Integrated Motion Analysis Suite (IMAS), which leverages multimodal data (clinical, sensor, and neuroimaging inputs) and multimodal machine learning (MML) to predict FM scores and motor recovery in stroke. Its potential is demonstrated via analysis of 28 stroke patients in acute and subacute phases of recovery, where features extracted from a set of motor tasks were used to predict FM scores and motor recovery, achieving a coefficient of determination (R^2) of 0.75 and Mean Absolute Error (MAE) of 2.83 and $R^2 = 0.83$ and MAE = 2.6 %, respectively.

IMAS is designed to continuously improve through its integration with a Big Data database, allowing for ongoing refinement of predictive algorithms as new data is collected in real-world clinical environments. Its ability to complement inherently limited clinical scales, handle incomplete data, and adapt to diverse applications highlights its potential for broader use in recovery after stroke, including long-term monitoring and precision rehabilitation.

Keywords

Stroke; Multimodal; Sensors; Machine Learning; Real World; Rehabilitation

1. Introduction

1.1. Challenges in stroke assessment and rehabilitation

Stroke is a leading cause of disability, affecting approximately 100 million worldwide [1]. Most survivors suffer motor deficits (~70 %) and require rehabilitation [2]. Effective rehabilitation, both near and long-term, should be guided by the patient's neurological examination (e.g., motor status) and potential for recovery, which is usually informed by clinicians' expert opinions. However, these impressions are influenced by personal experiences and cognitive biases, which explains the failure to accurately prognosticate outcomes even within the stroke specialist group [3]. In addition, current tools to predict the extent of a patient's motor recovery or even accurately measure their current abilities remain limited. Survivors undergo clinical evaluations with a narrow focus and thus do not receive comprehensive assessments before discharge. Tele-stroke sessions may partially compensate for such issues, yet they are limited in scope of care and are neither homogenous nor yet fully developed for functional assessments [4]. This is critical because "*all patients should undergo a formal assessment of the patient's rehabilitation needs before discharge*" [5]. With an aging population, a significant shortfall in US neurologists projected over the next 10 years [6], and a lack of neurologists outside of metropolitan areas [7], the challenge of providing timely and accurate assessments is growing.

Traditional clinical assessments such as the National Institute of Health Stroke Scale (NIHSS – list of acronyms is provided in Supplementary Material S1) and the Fugl-Meyer (FM) scale are widely used [8,9], and may be reduced to even coarser prognostic scales (e.g., Orpington Prognostic Scale (OPS)) [10]. These scales have notable limitations. They are highly dependent on the care provider/point of care, leading to intra- and inter-observer variability and influence of assessor experience. They often omit fractionated and complex distal movements, unequally weight upper versus lower limbs [11–13], and fail to address

real-world variability in patient conditions and environments, where motor function may differ significantly from what is observed during controlled assessments. The FM scale, while useful for comprehensively assessing post-stroke motor impairment, monitoring rehabilitation progress, and planning treatment, is subject to observer bias and inconsistency [9] and exhibits a ceiling effect in patients with mild motor impairment. Moreover, it does not account for the full spectrum of motor recovery, particularly in cases where finer motor skills and complex motor tasks are important.

The NIHSS begets inherent limitations as well [8]. For example, it sacrifices accuracy for reproducibility, under-represents certain stroke syndromes (i.e., posterior circulation, right-sided, infratentorial), lacks inter-rater reliability in some items, exhibits undesired redundancy in specific clinical contexts, presents cultural barriers, has a ceiling effect, and most importantly, like the FM, measures impairment and not disability. In summary, there is no single stroke scale that effectively captures all its complex effects [8]. Moreover, while nearly all acute stroke evaluations incorporate the NIHSS, rehabilitation trials typically use FM scores. This disconnect between acute care evaluations and rehabilitation metrics underscores the need for tools that holistically evaluate motor recovery, capturing both detailed motor performance and functional adaptability in diverse patient environments.

To address the aforementioned limitations the American Heart Association (AHA) Stroke Rehabilitation guidelines [14] specifically call for the development of “computer-adapted assessments for personalized and tailored interventions”, “newer technologies such as... body-worn sensors”, and “better predictor models to identify responders and non-responders”. However, user-friendly, clinician-tailored, patient-centered technology for a holistic real-world assessment and classification of motor symptoms and disability, as well as recovery prediction is lacking [15–17]. As elaborated on in Section 1.2, current solutions suffer from technical limitations and are often confined to controlled environments, as part of close-set data collection studies, which restricts their clinical applicability in real-world settings. Many of these tools rely on single-sensor modalities or fixed-point data collection methods, lacking the flexibility needed to handle the variability and noise commonly encountered in hospital settings. Thus, these systems fail to offer holistic, adaptable solutions capable of continuous monitoring, real-time assessments, and/or predictive modeling across diverse patient populations with varying degrees of stroke severity, comorbidities, recovery trajectories, and/or rehabilitation settings.

1.2. Integrated Motion Analysis Suite (IMAS): Multimodal Machine Learning (MML) for stroke motor assessment and recovery

Neuroimaging, wearable and portable sensors for measuring movement kinematics and kinetics, and integrated technology approaches are increasingly used in stroke clinical assessment [18], clinical scales prediction (e.g., FM), or motor recovery prediction [15,19]. While the prognostic value of neuroimaging markers (e.g., Diffusion Tensor Imaging (DTI) measures of white matter integrity) have been highlighted, current markers cannot fully assess the functional status of preserved tissue (and thus functional outcomes) [20]. Furthermore, methods such as [15,21] that combine additional functional assessments with imaging are not widely adopted potentially due to limitations such as risks and safety

concerns [22]. Similarly, sensor-based kinematic and kinetic measurements are being more commonly used to quantify motor abilities (e.g., [18,23–26]). However, many sensor technologies, such as those embedded in rehabilitation robots or complex motion capture systems, are not easily deployed in acute bedside settings due to size, cost, or complexity [27]. Wearable and portable systems have shown promise [18]. Unfortunately, though, current systems largely rely on a single sensor modality or focus on specific joints and/or symptoms, limiting their ability to capture the full complexity of motor behavior and disease state [18].

The IMAS used in this study (see detailed description in Section 2.2) integrates multimodal sensors for recording of movement kinematics and kinetics, and specifically inertial sensors, a portable camera-based system, and a portable force plate. In addition to data from these sensors, IMAS is designed to incorporate clinical information such as patient demographics, time from stroke onset, clinical assessment data, and where available, neuroimaging data (e.g., infarct volume) to derive a comprehensive assessment of patient’s motor status and potential for motor recovery.

The sensor types integrated into the IMAS have all been used separately for motor assessments in stroke, but all have specific limitations when used separately [28]. For example, inertial sensors provide only acceleration data along specific axes, with velocity and displacement estimates hindered by drift errors and offset fluctuations [29]; force plates are unable to directly measure joint position or coordination patterns [30]; and cameras are constrained by line-of-sight occlusion, sensitivity to reflections and background noise [31]. This limits its standalone utility for comprehensive motor assessment. Ultimately, systems based on a single sensor/modality have inherent limitations.

These limitations can be addressed by capturing and integrating data from multiple sensor types. A holistic view of motor function may be rendered by recordings of joint positions, movements, and coordination across the motor system. In addition, this raw data from different modalities must be synchronized, aligned, fused into a cohesive representation, and properly analyzed to prevent data fragmentation and subsequent limitations in the evaluation of a patient’s motor status and prediction of recovery potential. IMAS addresses this challenge by applying MML techniques to maximize the utility of these data types.

First, the system employs feature-extraction methods to derive metrics descriptive of upper limb movements (e.g., acceleration-based metrics from inertial sensors and position and velocity metrics from cameras). Next, IMAS uses modality fusion techniques to align and combine these metrics. Following fusion, dimensionality is assessed using Principal Component Analysis (PCA). Finally, elastic net is used both for feature selection, refining the model by eliminating less relevant features and for predicting motor impairment and motor recovery, focusing on the most predictive metrics. Herein, the system was explored using the sensor-based kinematic measures alone and coupled with the neuroimaging measures. We hypothesized that this multimodal approach, which integrates kinematic measures from various sensor types and neuroimaging data, could predict FM scores and motor recovery, contributing to the advancement of data-driven methods for stroke rehabilitation research.

1.3. Novelty of the research

The key contributions of this work are:

- Integrated multimodal framework: IMAS fuses clinical, wearable-sensor, and neuroimaging data via MML to predict both baseline motor impairment (FM scores) and subsequent recovery.
- Interpretability: The system identifies specific sensor-derived metrics linked to recovery outcomes, enabling clinicians to tailor therapy to individual impairments.
- Scalability: IMAS is designed to evolve continuously through integration with Big Data.
- Real-world readiness: The features of portability, minimal setup time, and data fusion methods that tolerate incomplete or noisy data, make IMAS suitable for diverse clinical environments.

1.4. Research significance

Stroke remains a leading cause of long-term motor disability, yet standard assessments suffer from bias, ceiling effects, and/or limited real-world applicability. By enabling the integration of high-dimensional, multimodal data into clinically actionable insights, IMAS offers a new path toward objective, scalable, and precision rehabilitation. Its potential for deployment in diverse environments (from bedside to telehealth) supports broader goals of health system modernization, outcome tracking, and more equitable access to post-stroke rehabilitation.

The remainder of this paper is organized as follows. Section 2 describes the data collection procedure and experimental protocol. It then describes data pre-processing steps, including data fusion and reduction, and prediction algorithms. Section 3 presents the results, Section 4 provides the discussion, and Section 5 the conclusions.

2. Study design

2.1. Subjects

The inclusion criteria were as follows: Adults (18–85 years old), able to provide informed consent, with a clinical presentation and neuroimaging (computed tomography angiography (CTA) – CT perfusion (CTP) / magnetic resonance imaging (MRI) – magnetic resonance angiography (MRA)) consistent with the diagnosis of acute ischemic or hemorrhagic stroke; Preserved mental status (Glasgow Coma Score > 12; Eye Response (4), Verbal Response (5), Motor Response (4–6)); Presence of upper limb weakness according to NIHSS (1–2 points in the arm) and ability to perform testing (i.e., NIHSS motor score 1–2 at elbow, wrist, and finger flexion–extension) within 30 days from stroke; or 2 points in the arm and ability to perform testing (i.e., NIHSS motor score 2 at elbow, wrist, and finger flexion–extension) in subacute stroke; Baseline Modified Rankin score < 4.

The exclusion criteria were as follows: History of dementia, as reported by a relative or noted in medical records; Presence of receptive aphasia at baseline or after the current acute

stroke; Need for rapid clinical response due to conditions such as psychosis, or suicidality; Unstable medical conditions (e.g., uncontrolled diabetes, uncompensated cardiac issues, heart failure, pulmonary issues, or chronic obstructive pulmonary disease); Amputated limbs; Absence of weakness on NIHSS (0 points = no drift for motor arm and leg items) or severe motor impairment (NIHSS 4 points for motor arm); Stroke mimics (e.g., infections, medication side effects, electrolyte imbalances); Worsening stroke symptoms between assessments.

All subjects volunteered for the study and gave their written informed consent. The experimental protocol was approved by the Institutional Review Boards of Case Western Reserve University (Cleveland, OH), Louis Stokes Veterans Affairs Medical Center (Cleveland, OH), the University of Illinois at Chicago (Chicago, IL), and OSF Health Care Little Company of Mary (Evergreen Park, IL). The study [clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT05469438) registration number is [NCT05469438](https://clinicaltrials.gov/ct2/show/study/NCT05469438). N = 30 stroke patients were enrolled in this study. Two of the study's 30 enrolled patients were excluded from the analysis herein as one completed less than half of the planned IMAS tests on the 1st evaluation (the patient suffered from uncontrolled rheumatoid arthritis) and the second developed a urinary tract infection during the admission which confounded the neurological exam due to altered mental status.

2.2. Apparatus

The IMAS used in this study is shown in Fig. 1 and was previously described in [28,32]. Briefly, it included a portable camera-based system (sampling frequency, $f_s = 30$ Hz), wearable three-axis gyroscope/three-axis accelerometer inertial measurement units (IMU) ($f_s = 64$ Hz), a portable force plate ($f_s = 98$ Hz), and a remote control for event marking (not shown in Fig. 1). The camera system included an embedded infrared sensor for measuring depth, i.e., 3D recordings, and software for segmenting the human body from the background, modeling the body as a 20-joint skeleton (hip center, spine, shoulder center, head, left and right shoulders, elbows, wrists, hands, hips, knees, feet, and ankle joints), and tracking 3D positions of the 20 joints (indicated as blue circles in Fig. 1). The standing force plate was not implemented in this study (all patients were evaluated while seated).

2.3. Experimental protocol

The experimental protocol consisted of up to five evaluation sessions, with the first four sessions being planned within 30 days of stroke onset and a 5th session planned for ~ 6–8 weeks post-stroke onset, following completion of a rehabilitation program (independent of our study, as the testing was entirely observational, with no intervention provided or prospectively assigned as part of the study). For this analysis, we focused on the first and last evaluation sessions available for the patients.

Motor assessments of the impaired limb were performed with the FM but were augmented with a set of single-joint and multi-joint upper limb motor tasks including 1) elbow flexion/extension, 2) shoulder abduction/adduction, 3) wrist flexion/extension, 4) wrist abduction/adduction (radial deviation/ulnar deviation), 5) shoulder flexion/extension, and 6) finger flexion/extension. Observations from these tasks were used to assess FM scores for the corresponding movements, ensuring consistency with the FM scale. Standard FM

assessments covering items outside these tasks were performed as per established FM procedures. Reflexes and coordination were assessed during the neurological evaluation. A detailed description of the tasks is reported in Supplementary Material S2. The task movements were each repeated 10 times at a comfortable speed, with a 3 s rest between repetitions.

During the IMAS assessments, the IMUs were attached to the subject's body with Velcro straps or elastic cloth material that cuffed the body segment, with anatomical landmarks guiding the positioning. For example, for tasks 1–6, the primary IMU's placement was as follows: top side of the patient's wrist for task 1, back of the hand for tasks 3 and 4, close to the elbow, between the shoulder and elbow, for tasks 2 and 5, and on the index finger, proximal phalanx for task 6. For the FM evaluations, the primary IMU's placement was on the wrist. Our analysis focused on the within synergies and the hand-to-lumbar spine (part of mixed synergies) assessments.

The IMUs location and order of tests were chosen as a trade-off between time maximizing completeness of information, minimizing overall IMAS-testing duration and number of changes of sensor configuration. All tests were tracked with the camera system. Throughout the experiments, the camera and subject chair were kept in fixed positions to minimize set-up times between sessions, prevent errors due to equipment re-positioning, and maintain consistency between participants. The remote control allowed the experimenter to mark recordings; the marker signal was set whenever an event occurred (e.g., beginning or end of each motor task or, for motor tasks involving multiple repetitions, of each single movement) and was null otherwise.

In addition to the FM assessments (focused on the Upper Extremity (UE), where higher scores indicate less motor impairment), OPS, Barthel Index (BI), and NIHSS assessments were also planned for the 1st and 5th evaluation. The OPS is a prognostic tool (lower scores indicate a greater likelihood of recovery); the BI measures functional independence (higher values reflect greater independence); and the NIHSS evaluates neurological impairment (higher scores denote greater severity).

These measures were performed to complement the motor assessment data and provide broader functional and prognostic insights. This set of assessments was chosen to balance the robustness of data with feasibility (i.e., to avoid patient fatigue). As part of the patients' 1st evaluation, clinical data was also collected, including stroke lesion information (from CT/CTA and MRI/MRA (e.g., infarct volume and location)), demographics, and time from stroke onset. Standard neurological examinations performed by a neurologist were also collected. Finally, as the study took place during the COVID-19 pandemic, patients unable to return to the laboratory for their final visit had their final FM determined via the patients' medical record by a neurologist at the collaborating hospitals, using the method of Chhatbar et al. [33] or via the last available FM evaluation recorded in our laboratory (the latest date available).

2.4. Data analysis

2.4.1. Pre-processing and data fusion: Prior to each recording session, all sensors underwent a brief calibration routine to verify correct sampling and proper placement. For each motor task, data was recorded and saved in separate files, one for each sensor. Synchronization was achieved using the remote-control marker as a reference signal, ensuring temporal alignment across all sensors.

For all movements, speed profiles v were calculated from the first-order derivative of the 3D movement trajectories recorded with the camera and smoothed with a 10 Hz low-pass FIR filter, similar to [26,28]. Angular velocity signals from the gyroscope (X_{rot} , Y_{rot} , Z_{rot}) were filtered with a 4th-order low-pass Butterworth filter (5 Hz cut-off). Since sensors sampled data at different frequencies, resampling was applied to standardize the data across modalities. For each test, the signal corresponding to the performance of the motor task was extracted using the first and last occurrence of the remote-control marker signal. For motor tasks involving multiple movements, individual movement segments were extracted using marker signals, which identified the start and end points of each movement. Following segmentation by markers, the algorithm applied peak detection on the gyroscope's Z_{rot} signal, iteratively adjusting the threshold from 0.055 up to 0.4 times the maximum signal value. This thresholding identified points where the signal crossed below the threshold on both sides of each detected peak, isolating individual movement profiles. These points were used to segment both IMU and camera data, and in cases of ambiguities, such as unclear stopping times, a visual comparison of signals from multiple sensors (IMU-derived thresholds vs. minimums of camera-derived v signals) was employed to resolve discrepancies. After aligning all data and addressing ambiguities, the sensor data were integrated into a comprehensive dataset. This fused dataset was then used for further analysis, including feature extraction, dimensionality reduction, and prediction, as detailed below. Custom C# routines were written to synchronize the recordings from all the IMAS sensors and the remote control, while custom MATLAB routines were used for data analysis. Missing data was imputed using a mean imputation Intention-to-Treat (ITT) approach [34].

2.4.2. Feature extraction: IMAS metrics were extracted from the signals recorded with the IMAS segmented as above as follows. For camera data, analysis focused on a joint or body segment of interest, selected based on the specific task (task 1 and the FM evaluation tasks: wrist; tasks 3 and 4: hand; tasks 2 and 5: elbow). For each segment, movement metrics were calculated, including mean speed, maximum speed, duration, smoothness (ratio of mean speed to maximum speed), number of movements, and path length traveled by the joint or body segment of interest, similar to [28]. For IMU data, mean and maximum rotational speed magnitudes were calculated. For task 6, where camera data was not applicable, mean and maximum acceleration magnitudes were also computed. For tasks repeated multiple times, mean and standard deviation values were calculated across the segmented data portions, separately for the flexion/extension or abduction/adduction phases of the movement.

Although not part of the IMAS system, imaging data provided structural insights to complement sensor-based assessments. Imaging metrics were calculated by a radiologist. For ischemic stroke, lesion volume was calculated as described in [35], starting from area measurements obtained via the manual tracing method [36]. The infarct area was measured on the first available MRI scan for each subject, focusing on the Diffusion-Weighted Imaging (DWI) (b-1000) sequence to identify regions of restricted diffusion indicative of ischemic injury. High-signal intensity regions (visually perceptible bright areas) were manually delineated as Regions of Interest (ROIs) using a closed polygon tool, with measurements performed slice-by-slice in the craniocaudal direction (from the uppermost to the lowest axial slice). For hemorrhagic stroke, lesion volume was calculated using the ABC/2 formula [37]. Additionally, the Alberta Stroke Program Early CT Score (ASPECTS) was recorded to assess early ischemic changes [38]. Imaging data were processed using RadiAnt DICOM Viewer 2022.1 or the GE ZFP Viewer integrated into the hospital's electronic medical record (EMR) System.

2.4.3. Data reduction: PCA was used to analyze the correlation structure and estimate the effective dimensionality of each dataset. Each measure in each data set was standardized by removing its mean and dividing by its standard deviation and PCA analyses were conducted for the set of FM and IMAS measures, as well as for the NIHSS, OPS, BI, and imaging measures, separately and together.

2.4.4. Prediction algorithms: Algorithms were developed to predict: 1) FM scores (using IMAS and imaging metrics from the first evaluation to predict FM scores at the same evaluation); and 2) motor recovery (using IMAS and imaging metrics from the first evaluation to predict recovery outcomes, as measured by FM determined directly or indirectly (e.g., from EMR) as described above, at the final evaluation, normalized from 0–100 % (based on the last FM evaluation maximum scale). Additionally, algorithms were developed for predicting the following exploratory outcomes: 3) BI and OPS scores (using IMAS and imaging metrics from the first evaluation to predict scores at the same evaluation), and 4) relative recovery outcomes defined as (normalized FM at the final evaluation – normalized FM at the first evaluation)/ (normalized FM at the first evaluation).

For FM, BI, and OPS predictions, models used data from within the same evaluation (IMAS data collected at the first evaluation to predict scores at the same time point). For recovery prediction, initial evaluation data were used to predict final recovery, aligning with real-world rehabilitation scenarios where baseline data guide long-term planning.

Prediction algorithms used a sparse linear model fit by elastic net optimization [39]. Elastic net regularization [39] is a well-established approach for regularization and variable selection for models for which a sparse subset of a large number of candidate variables are thought to provide the preponderance of information for prediction and these candidate variables include groups of co-varying predictors that should be included together in the model. This approach allowed us to identify which features mattered most for each prediction. The elastic net penalty parameters were systematically varied using a grid search approach to find the best combination of predictors with highest R^2 value with the model Degrees of Freedom (DFs) capped at 50 % of the patient group size. The number of DFs

was capped to further reduce model complexity, similar to our earlier study [28]. Model performance was assessed with Leave-One-Out Cross-Validation (LOOCV) [40] to evaluate the model's predictive accuracy, quantified primarily by R^2 and Mean Absolute Error (MAE) metrics. Additional metrics (Variance Accounted For (VAF), Root Mean Square Error (RMSE), and the Ratio of RMSE to the Standard Deviation of the Observations (RSR)) were also computed for completeness.

3. Results

3.1. Participant characteristics and data collection

The 28 assessed participants' characteristics, including demographics and evaluation timing, are summarized in Table 1. Lesion types included 23 ischemic, 4 hemorrhagic, and 1 combined case. Lesion involvement was categorized as follows: motor cortex involvement (5); motor thalamic involvement (3); white matter involvement affecting regions such as the posterior limb of the internal capsule, cerebral peduncle, brainstem, or corona radiata (25); and/or other areas (19) where most subjects had multiple areas involved. The average UE FM motor function score at the first visit in our patient sample was 50 ± 8.1 . 8 patients' final FM scores were determined from medical records as above at their last stroke clinic visit, 16 via their FM evaluations at our clinic (9 at the 5th evaluation, 7 from an earlier evaluation), and 4 patients' first and last FM evaluations were on the same day (3/4 had first evaluation FM scores within 90 % of the maximum and their inclusion/exclusion in the final data set had minimal impact on recovery prediction results (see below)). Fig. 2 shows example accelerometer data from two patients with different FM scores.

3.2. PCA analysis

A total of 197 features were used from the IMAS data per evaluation visit. Fig. 3 displays the PCA results based on the 1st evaluation. Although clinicians typically rely on a single clinical score to describe deficits in stroke patients, these results provide evidence that there is additional variability in the IMAS assessments beyond that explained by simple clinical scales alone. The light blue line shows the IMAS testing results, while the darker blue line shows the PCA results when the FM, IMAS, and other clinical measures are combined. The other lines show the PCA results when the individual clinical scales are used (e.g., FM). The plot of variance explained as a function of dimensions for the combined dataset is consistently close to that of the IMAS alone.

This indicates that adding IMAS data increases the number of independent measures beyond what is captured by FM and other clinical data. However, adding the FM data may not increase the number of independent measures beyond those provided by the IMAS data.

3.3. IMAS-based predictions

Results of the prediction models are summarized in Table 2 and Fig. 4; see Section 4 for a discussion on the clinical significance of the accuracy values. Fig. 4 presents the results of an optimized model for predicting FM scores from IMAS metrics, where both data sets were recorded during the first observation, achieving a mean prediction error of 2.82 points across the patient group. This is compared to the standard deviation of FM scores, which is 8.11

points ($R^2 = 0.75$). The mean FM total score at this observation is ~50. This result indicates that IMAS metrics are effective in predicting FM scores. As for FM scores prediction, adding Imaging metrics to the IMAS metrics did not improve the model performance nor change the predictors. These results suggest that IMAS metrics can be sufficiently robust for predicting FM scores under the DF constraint and Imaging metrics, in this context, did not add unique predictive value. The model for predicting BI scores from IMAS metrics reached $R^2 = 0.56$ and MAE of 8.5 points across the patient group, compared to the standard deviation of 15.81 of the BI scores. Similarly, the model for predicting the OPS scores from IMAS metrics reached $R^2 = 0.58$ and MAE = 0.46 across the patient group, compared to the standard deviation of 0.86 of the OPS scores. Adding Imaging metrics did not change these models' performance. The superior performance of IMAS metrics in predicting UE FM scores compared to the OPS and BI scores is expected given the closer alignment of IMAS metrics with the FM scale's focus on motor-specific impairments, whereas OPS and BI also incorporate balance, cognition, and broader functional tasks, making them less directly tied to the movement-specific metrics emphasized by IMAS.

Recovery predictions at the last evaluation, from IMAS metrics extracted from data recorded during the first evaluation, achieved an MAE of 2.6 % across the patient group, compared to the standard deviation of 7.7 % ($R^2 = 0.83$). This result supports the premise that IMAS data from the first evaluation may be used to predict recovery and further supports the standalone strength of IMAS metrics in capturing motor impairments. As for recovery prediction, differently from the prediction of 1st evaluation FM scores, adding imaging data improved the model performance ($R^2 = 0.89$ and MAE = 2 %). A partial F-test comparing the IMAS-only ($R^2 = 0.83$) and IMAS + imaging ($R^2 = 0.89$) recovery models yielded $F(1,13) = 6.2$, $p = 0.027$, indicating a statistically significant improvement when including imaging. This corresponds to a 0.6 % reduction in MAE (from 2.6 % to 2.0 %). This result suggests a complementary value of combining sensor-based and imaging data for prediction of recovery outcome, highlighting the potential of multimodal integration for more precise outcome prediction.

Relative recovery was analyzed as an exploratory outcome. Using IMAS metrics collected during the first evaluation, the prediction model achieved an $R^2 = 0.74$ and an MAE of 8.2 %. This indicates the model effectively captured variability in relative recovery outcomes, with an average error close to the observed mean recovery (Mean = 7.7 %, SD = 22.4 %) in the analyzed data set.

3.4. Predictors for FM and recovery

Predictors of each model are detailed in Table 2. For FM prediction, notable metrics included *IMAS_EL_RDS* (path length during elbow extension, $\beta = -0.11$), indicating penalties for inefficient or inconsistent gross motor movements, and *IMAS_EL_RStdDuration* (variability in elbow extension duration, $\beta = -0.23$), which highlights the importance of temporal consistency. The inclusion of *IMAS_FI_Number* (number of finger flexion/extension movements, $\beta = 0.21$) highlights the contribution of finer motor tasks, such as finger flexion/extension, to the FM scale, complementing its broader emphasis on gross motor control. Additional predictors, such as

IMAS_SY_LMeanRot (mean rotational speed during the flexion part of the within-synergy FM task, $\beta = 0.15$), reflect faster movement execution associated with higher FM scores, potentially indicating more efficient task performance. Variability metrics like *IMAS_SE_RStdMeanRot* (variability in mean rotational speed during shoulder extension, $\beta = 0.10$) suggest controlled variability as a key aspect of movement consistency. Furthermore, *IMAS_WR_LDuration* (wrist flexion duration, $\beta = -0.18$) highlights the importance of efficient timing in distal movements. These findings, derived from a small dataset, illustrate IMAS's ability to capture diverse aspects of motor function relevant to the FM scale.

For recovery outcomes prediction from IMAS-only metrics, key predictors included *IMAS_SE_RDS* (shoulder extension path length, $\beta = -0.36$), which reflects inefficiency or instability in gross motor movements, and *IMAS_SY_LMeanRot* (mean rotational speed during the flexion part of the within-synergy FM task, $\beta = 0.29$), reflecting faster movement execution that may be indicative of improved coordination or task efficiency. Fine motor predictors, such as *IMAS_WR_RDS* (wrist extension path length, $\beta = 0.10$) and *IMAS_WR_LMeanRot* (mean rotational speed during wrist flexion, $\beta = -0.30$), highlight the importance of precise distal movements. Variability metrics like *IMAS_SA_LStdMeanSpeed* (variability in mean speed during shoulder abduction, $\beta = -0.24$) emphasize the value of consistent motor performance. These results suggest that IMAS metrics can capture recovery-specific predictors, but further validation in larger datasets is needed.

When imaging metrics were incorporated into the recovery model, the structural metric *IM_MRI_ischemic_vol* (lesion volume, $\beta = -0.07$) was retained, indicating that larger lesions are associated with poorer recovery outcomes in our patient population. Movement performance metrics like *IMAS_SE_RDS* (shoulder extension path length, $\beta = -0.33$) and *IMAS_SY_LMeanRot* (mean rotational speed during the flexion synergy of the within-synergy FM task, $\beta = 0.33$) remained consistent with the IMAS-only model, reinforcing their importance in recovery prediction. These findings underscore the potential complementary role of structural and functional data in enhancing predictive accuracy.

3.5. Exploratory analysis

IMAS metrics were also evaluated for their ability to predict broader outcomes such as BI (functional independence) and OPS (prognostic) scores, as well as relative recovery. For BI prediction, gross motor metrics such as *IMAS_EL_LDS* (elbow flexion path length, $\beta = -0.28$) were prominent, along with variability metrics like *IMAS_SY_LDuration* (duration of the flexion part of the within synergy FM task, $\beta = -0.09$). OPS predictors included *IMAS_SA_LDS* (shoulder abduction path length, $\beta = -0.24$) and *IMAS_SE_LStdDuration* (variability in shoulder flexion duration, $\beta = 0.39$).

Relative recovery predictors included *IMAS_EL_RDuration* (elbow extension duration, $\beta = 0.28$) and *IMAS_WR_LDuration* (wrist flexion duration, $\beta = 0.25$), *IMAS_FI_Number* (number of finger flexion/extension movements, $\beta = -0.27$), and *IMAS_SA_RStdSmoothness* (variability in shoulder adduction smoothness, $\beta = 0.14$). Imaging metrics did not contribute significantly to this model, suggesting that IMAS metrics alone captured the key dimensions of relative recovery in this dataset.

To explore the impact of the 4 patients whose first and last evaluation visits were done on the same day, we completed the same analysis with the patients excluded from the dataset and the above predictors were still important predictors in the final recovery models (absolute and relative).

4. Discussion

We presented a system that integrates multimodal data with MML techniques to address the complexities of stroke assessment and recovery prediction. By incorporating diverse data sources, managing uncertainties, and accommodating incomplete datasets, IMAS provides a robust framework for analyzing and predicting motor outcomes, bridging the gap between real-world clinical variability and the need for scalable, data-driven rehabilitation tools. Using data from 28 acute and subacute stroke patients, we demonstrated the potential of IMAS. IMAS metrics exhibited variability beyond that captured by traditional clinical scales, underscoring their added value. They predicted FM scores ($R^2 = 0.75$, MAE = 2.82) and motor recovery ($R^2 = 0.83$, MAE = 2.6 %), with further improvement when neuroimaging metrics were incorporated ($R^2 = 0.89$, MAE = 2 %). Beyond FM and recovery, IMAS metrics predicted BI and OPS scores, albeit with lower performance, as well as relative motor recovery. The predictors varied across contexts, reflecting a dynamic balance between gross and fine motor abilities depending on the specific outcome, highlighting IMAS's versatility and adaptability to diverse recovery scenarios.

To ensure these predictive strengths translate into robust performance in noisy, incomplete, or variable hospital environments, IMAS incorporates several safeguards. IMAS synchronizes each sensor's output using a common marker supplied by the remote control, permitting precise alignment (or re-synchronization) of all data streams even when individual samples are lost. Moreover, routine per-session calibration verifies that each sensor maintains its intended sampling rate. These measures safeguard data integrity and explain why IMAS maintains predictive accuracy in real-world clinical settings, where noise, hardware variability, and incomplete recordings are inevitable.

Clinically, our work addresses a *critical knowledge gap*: the evaluation of motor recovery in patients with acute/subacute mild stroke [41]. These patients experience subtle yet meaningful motor deficits (e.g., reduced fine motor control, fatigue, difficulty with limb coordination, etc.), which frequently limit performance in complex real-world tasks, decrease mood and quality of life [42,43], and limit performance or impede the return to work [44]. Yet, these deficits are poorly captured by scales such as the FM or NIHSS assessments which suffer from a lack of sensitivity in capturing more subtle or complex motor impairments and ceiling effects in populations with mild deficits. While sensor-based and neuroimaging approaches offer greater objectivity, they are typically designed for controlled research environments and struggle with the variability, noise, and incompleteness that characterize real-world clinical data. Furthermore, the increasing use of automated data collection in healthcare environments has introduced additional challenges, including the management of large volumes of data that frequently contain missing or incomplete values. Addressing this complexity requires robust approaches capable of integrating disparate data sources and maintaining performance amidst variability and noise.

IMAS addresses these limitations by integrating clinical assessments with sensor-based metrics and neuroimaging data into a comprehensive framework for predicting patient motor status and recovery. By leveraging sensor-based data recorded during various motor tasks, IMAS provides a more nuanced and comprehensive motor status view, which aligns with real-world rehabilitation needs. Our dataset, collected from patients in a hospital setting, captures the inherent variability of clinical environments and practical challenges, such as variable patient recovery trajectories and incomplete data. While FM scores were the primary focus for predicting motor recovery, only a subset of patients had complete FM assessments at the final evaluation, necessitating the use of alternative data sources such as hospital records to fill these gaps. We benefitted from the strong correlation of the UE subsets of NIHSS and FM scale [33] to approximate FM outcomes effectively and maintain predictive accuracy. This adaptability underscores IMAS's robustness in handling real-world data challenges, demonstrating its capacity to integrate disparate sources and produce accurate predictions even in the presence of incomplete datasets.

4.1. Dimensionality analysis and feature selection

PCA was used to analyze the dimensionality of the IMAS dataset, revealing the additional complexity and independent information provided by IMAS metrics compared to traditional clinical scales. The analysis highlighted the complexity of IMAS data, which captures variability and subtleties in motor function that clinical scales alone cannot. Retaining diverse and informative features is critical for addressing real-world variability in patient conditions and data quality.

The prediction capabilities of IMAS were further demonstrated using a sparse linear model guided by elastic net optimization. This approach balances L1 and L2 regularization, ensuring model interpretability while reducing the risk of overfitting, a common challenge when working with high-dimensional datasets. The elastic net model systematically identified the most relevant features for predicting FM scores and recovery outcomes. This MML approach ensures that features from diverse data sources are effectively integrated, facilitating the prediction of both the motor impairments reflected in FM scores and the improvements associated with recovery outcomes.

While some predictors, such as controlled variability metrics, are shared between these two models due to their influence on general motor performance, others are uniquely aligned with either FM scores, which emphasize task-specific precision, or recovery outcomes, which prioritize broader functional adaptability by integrating improvements in both gross and fine motor control.

4.2. Advancing stroke outcome and recovery prediction using IMAS metrics

Compared to existing stroke recovery prediction models, such as the Predict Recovery Potential (PREP) algorithm [15] and Rehabilomics [45], IMAS has several advantages. Both PREP and Rehabilomics primarily rely on specific neuroimaging markers or single data sources, which limit their adaptability to the complexity and variability of real-world clinical settings. For instance, PREP uses motor evoked potentials ((MEPs), a measure that requires sophisticated equipment and skillset to administer) and DTI data (usually

not part of clinical imaging set) to predict upper limb recovery but lacks the ability to handle noisy or incomplete data, a common issue in acute clinical environments. Similarly, Rehabilomics integrates biological and environmental factors to predict long-term recovery but focuses primarily on neuroimaging data, offering limited flexibility for integrating continuous sensor-based inputs or real-time motor assessments.

IMAS, in contrast, integrates multimodal data sources through data fusion techniques, making it adaptable to incomplete or noisy datasets. The identified predictors for FM and recovery outcomes highlight IMAS's potential to address distinct aspects of stroke recovery: task-specific motor performance and broader functional adaptability. For FM prediction, metrics such as *IMAS_EL_RDS* (path length during elbow extension, $\beta = -0.11$) and *IMAS_SA_RStdMaxSpeed* (variability in maximum speed during shoulder adduction, $\beta = -0.11$) align with the FM scale's emphasis on precision and consistency in structured motor tasks. Fine motor metrics like *IMAS_FI_Number* (number of finger flexion/extension movements, $\beta = 0.21$) illustrate the supplementary role of dexterity in FM scores, while gross motor metrics such as *IMAS_SY_LMeanRot* (mean rotational speed during the flexion part of the within-synergy FM task, $\beta = 0.15$) highlight the FM scale's sensitivity to impairments in gross motor task performance during structured tasks. For recovery prediction, broader functional adaptability is reflected in metrics like *IMAS_SE_RDS* (shoulder extension path length, $\beta = -0.36$) and *IMAS_SY_LMeanRot* (mean rotational speed during the flexion part of the within-synergy FM task, $\beta = 0.29$). Wrist movements, such as *IMAS_WR_LMeanRot* (mean rotational speed during wrist flexion, $\beta = -0.30$), play a crucial role in bridging gross and fine motor control, underscoring the need for coordination and precision in recovery. These findings, though preliminary, demonstrate IMAS's ability to capture diverse dimensions of motor performance and recovery. This predictive granularity supports the development of precision rehabilitation strategies, in which therapy intensity and focus can be adjusted based on a patient's motor impairment metrics. However, given the small sample size, the specific role of these predictors requires further validation in larger and more diverse datasets.

The predictive performance of IMAS also supports its application in guiding precision rehabilitation. An R^2 of 0.75 indicates that 75 % of inter-patient variability in baseline FM scores is captured by IMAS features, and an MAE of 2.8 points (cohort standard deviation (SD) ~ 8) is well below the half-SD threshold (~ 4 points) commonly used as a minimal clinically important difference [46]. In our sample (where the SD of raw FM change was ~ 5), a 3–4-point gain likely reflects true, patient-perceived improvement. Thus, an MAE of ~ 2.8 points suggest IMAS can reliably stratify patients (mild vs. severe), for example at baseline to guide early therapy intensity: those predicted mildly impaired might proceed to task-oriented or moderate-intensity exercises, whereas those predicted severely impaired might be steered toward compensatory strategies or supportive technologies. Similarly, for recovery prediction, an R^2 of 0.83 and MAE of 2.6 % (SD = 7.7 %) indicate that IMAS can forecast relative gains with sufficient precision to inform rehabilitation planning, for example, directing patients with predicted low recovery toward compensatory training and those with high predicted gains toward more intensive therapies (e.g., robot-assisted therapy).

4.2.1. Imaging metrics and recovery: The inclusion of Imaging metrics in recovery prediction underscores the complementary role of structural imaging data in capturing recovery potential. For example, *IM_MRI_ischemic_vol* (lesion volume, $\beta = -0.07$) indicates that larger lesions are associated with poorer recovery outcomes, while functional predictors such as *IMAS_SE_RDS* (shoulder extension path length, $\beta = -0.33$) and *IMAS_SY_LMeanRot* (mean rotational speed during the flexion part of the within-synergy FM task, $\beta = 0.33$) remained consistent across models. The improvement in model performance (R^2 from 0.83 to 0.89) highlights the added value of integrating structural imaging data with sensor-based metrics, particularly in refining predictions and tailoring rehabilitation strategies. When imaging metrics are added, the MAE narrows from 2.6 % to 2.0 %. Clinically, even modest improvements in accuracy may be meaningful when patient outcomes fall near critical decision thresholds, for example, distinguishing between those expected to achieve moderate versus high recovery. In such cases, reducing prediction error may improve triage decisions and therapy planning.

Structural imaging topography has been shown to influence UE motor recovery [20]. In our recovery model, lesion volume (*IM_MRI_ischemic_vol*, $\beta = -0.07$) was the only retained imaging predictor. This may reflect characteristics of our cohort, where isolated small subcortical strokes (particularly ischemic events) were underrepresented. As a result, the model may overweight lesion volume as a predictor, since more subtle location-specific effects (e.g., small subcortical infarcts) were not common in the cohort. Additionally, among the few subcortical cases included, several were hemorrhagic rather than ischemic; these patients often exhibit different recovery trajectories, potentially contributing to the observed findings. Larger infarcts have long been linked to poorer functional outcomes and diminished responsiveness to standard rehabilitation protocols [20], implying a need for alternative or adjunctive therapies. Patients with larger lesions may benefit from pre-allocation to longer-term, high-supervision programs (e.g., inpatient or intensive day-hospital) to maximize recovery potential.

4.2.2. Exploratory outcomes: The exploratory analysis of OPS, BI, and relative recovery underscores the versatility of IMAS metrics to identify subtle, joint-specific differences across diverse clinical scales. For OPS, greater shoulder abduction path lengths were associated with better OPS scores. In contrast, for BI, longer elbow flexion path lengths reflected inefficient motor control and were associated with lower functional independence. For relative recovery (which provides a dynamic view of progress, whereas absolute recovery emphasizes overall outcomes), baseline slower or more variable movements predicted greater relative change while better baseline fine motor performance (finger flexion/extension) predicted smaller relative change. These findings, while promising, are exploratory and reinforce the need for larger studies to validate the identified predictors and refine predictive models.

4.3. Scalability and future applications

Given the projected increase in global stroke incidence [1], and the well-documented inaccuracy of clinicians to predict stroke outcomes [3], there is a pressing need for scalable, patient-centered, and clinician-friendly automated systems. Such platforms must not only

streamline existing workflows but also adapt as new therapies and diagnostics emerge. IMAS meets this need by offering a portable, low-cost platform which is minimally disruptive to routine clinical practice. As it uses wearable and portable sensors, it can be set up in either outpatient or inpatient environments and requires less than 5 min for full deployment and data collection. Training requirements are minimal, focused on proper sensor placement and basic system operation, making IMAS accessible to a broad range of healthcare personnel, including those without technical background. We have already demonstrated feasibility in Parkinson's disease (PD) [28] and chronic pain cohorts, underscoring the system's adaptability beyond stroke.

Seamless integration into hospital workflows is crucial in light of neurologist shortages, particularly in underserved regions [7]. IMAS leverages multimodal data to generate reliable evaluations despite noise or missing data, and its architecture supports Big Data integration, allowing refinement of predictions as more clinical and real-world data become available. Moreover, IMAS can interface with telehealth platforms to extend its reach to remote populations and support scalable home-based monitoring.

IMAS differs from traditional EMR-based models in three key ways. First, instead of relying solely on static clinical variables (e.g., labs, vitals, codes), IMAS fuses high-resolution kinematic and imaging inputs, synchronized via a common timestamp and filtered for noise, to extract movement metrics (e.g., speed, smoothness) that do not appear in routine charts. Second, IMAS's ITT approach can be set to prioritize each patient's own data (using within-patient means) before falling back on cohort averages. In contrast, many EMR pipelines immediately substitute a missing value with population means or discard incomplete records, introducing bias into the analysis. Finally, IMAS uses elastic-net regression to combine sensor, imaging, and demographic features into a sparse, interpretable model for predicting FM scores. These innovations together allow IMAS to handle real-time, noisy, and incomplete data more robustly than static EMR systems.

By offering a scalable, multimodal assessment framework, IMAS can support precision rehabilitation strategies tailored to individual patient profiles, ultimately improving patient outcomes and reducing the burden on healthcare providers. For example, knowing that a patient could improve a certain number of points on the FM scale may enable clinicians to further document the indication for rehabilitation while adjusting treatment intensity or recommend targeted therapies to optimize functional gains. A high-level IMAS implementation workflow is shown in Fig. 5, while a possible precision rehabilitation therapy workflow that supports personalized goal adaptation is shown in Supplementary Material S3. Using data recorded from patients with PD with a similar version of IMAS, we previously showed that a self-organizing map could cluster patients in terms of motor symptoms such as bradykinesia, tremor, and postural instability [28]. A similar approach could be used to cluster patients based on motor impairments after stroke onset and investigate the effects of different rehabilitation strategies. This clustering could offer valuable insights into the effects of different rehabilitation strategies, potentially revealing optimal approaches for various patient subgroups.

Future iterations of IMAS may incorporate continuous monitoring devices and in-hospital motion tracking systems, to explore the feasibility of capturing real-time data on patient movement and physiology. If successfully integrated, this approach could support IMAS monitoring of recovery during the critical early rehabilitation phase. Additionally, its modular architecture could enable integration with EMRs, facilitating the use of historical patient data for more comprehensive motor recovery predictions. As data from diverse patient groups and clinical environments is collected, IMAS's predictive algorithms will be refined. This continuous improvement is facilitated by IMAS's ability to incorporate real-time data from new clinical contexts, ensuring the system remains responsive to the evolving demands of stroke rehabilitation. Its capacity to handle incomplete or noisy data makes IMAS particularly suited for real-world clinical environments, where data may be inconsistently structured or vary in quality.

4.4. Limitations

This study has several limitations. First, while IMAS demonstrates robust prediction capabilities using multimodal data, the dataset is relatively small. This is a well-documented problem in the field, where clinical decision-making modeling is limited by overfitting. Nevertheless, we recruited a diverse population from different sites, which included > 50 % minority patients. Thereby, we believe our results herald promise for the generalizability of findings to broader population sets or varied rehabilitation settings. The identified predictors highlight IMAS's potential to capture diverse aspects of motor assessment and recovery; however, these findings remain exploratory. Moreover, we did not evaluate IMAS on an independent external validation dataset, which limits our ability to confirm its performance in truly unseen patient populations. Future work should involve larger, more diverse datasets from multiple sites, including external validation cohorts, prior to clinical translation to enhance scalability and applicability.

Second, this study focuses primarily on UE impairments and motor recovery, given that UE impairments constitute a major source of long-term disability (fewer patients regain UE function compared to lower-limb function [47,48] and because a well-defined set of kinematic features for UE movement is already established (see, e.g., [26])). Nonetheless comprehensive stroke rehabilitation requires evaluation of a broader spectrum of functional deficits, including lower-limb motor impairments and other domains such as cognitive or sensory deficits. Future developments of IMAS will integrate lower-limb kinematics and cognitive/sensory assessments (extensions that have already been piloted in other patient populations (e.g., PD [28]) to improve IMAS's utility for comprehensive rehabilitation assessments.

Third, reflexes and coordination were assessed during the neurological examination, but not as part of the FM scale due to participant lack of cooperation. This is a common conundrum in the real world, where it is well known that subjects' perception of poor performance discourages the use of the involved limb, measurably worsens motor execution, and may foster feelings of discomfort or shame [49].

Fourth, imaging metrics were limited to measures such as lesion volume. Including additional imaging data, such as lesion location or functional metrics (e.g.,

electroencephalography-based measures), could enhance prediction accuracy and further tailor rehabilitation strategies. Given IMAS's flexible design, future studies could focus on integrating these additional metrics to refine predictions and provide deeper insights into recovery potential.

Fifth, the recovery data set was limited due to the impact of the COVID-19 pandemic in comparison to the day 1 data sets used for the FM, BI and OPS predictions. While we used the data to demonstrate the applicability of the IMAS system, specific conclusions regarding the importance of individual predictors for the recovery assessments need to be validated in larger data sets.

Finally, the statistical approach we employed was based on linear models, which allow for unbounded predictions. We chose this method because the primary goal was to demonstrate the potential of the IMAS system for the prediction of clinical scales (e.g., FM) and outcomes and to explore appropriate predictors across these clinical scales and recovery modeling. While this approach provides models with interpretable components, more complex, non-linear techniques, such as deep learning, could potentially capture the complex relationships between IMAS data and clinical-scale outcomes more effectively. However, these methods often sacrifice interpretability and typically require larger datasets to achieve stable and generalizable performance. Future work will examine whether the trade-off between model flexibility and statistical efficiency justifies their use. We also plan to explore alternative modeling methods with bounded or discrete outcomes, such as multinomial generalized linear models and multilayer neural-network classifiers.

5. Conclusions

IMAS uses real-world clinical data and specifically multimodal, sensor-based data with optional neuroimaging data to predict FM scores and motor recovery potential. Our design and interface are geared toward patient comfort and ease of use. Although this study was conducted on a limited sample and lacks an independent validation cohort (rendering the reported predictive performance exploratory and necessitating confirmation in larger, diverse populations), the system's adaptability to handle incomplete or noisy data, coupled with its scalability for integration into larger clinical infrastructures, makes it a valuable tool for improving stroke rehabilitation outcomes. These ideas align with the need to place patients and physicians at the center of these developments given that each patient may define acceptable outcomes in a different way. We believe that our patient-centered, clinician-friendly approach has the potential to translate effectively into clinical practice. Our future work will focus on continuing our collaboration with clinicians to validate the system with larger datasets and expand its application to other neurological conditions.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Laura Dipietro reports financial support was provided by Highland Instruments, Inc. Tim Wagner reports financial support was provided by Highland Instruments, Inc. Paulo E. P. Teixeira reports financial support was provided by Highland Instruments, Inc. Laura Dipietro reports a relationship with Highland Instruments, Inc. that includes: employment, equity or stocks, and funding grants. Tim Wagner reports a relationship with Highland Instruments, Inc. that includes: board membership, employment, equity or stocks, and funding grants. Paulo E. P. Teixeira reports a relationship with Highland Instruments, Inc. that includes: employment. Laura Dipietro has patents pending to Highland Instruments, Inc. Tim Wagner has patents pending to Highland Instruments, Inc. Uri Eden has patents pending to Highland Instruments, Inc. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Clinical trial data will be uploaded to [ClinicalTrials.gov](https://clinicaltrials.gov). The dataset is proprietary but available on request from the corresponding author, subject to necessary approvals.

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Schematic of IMAS hardware

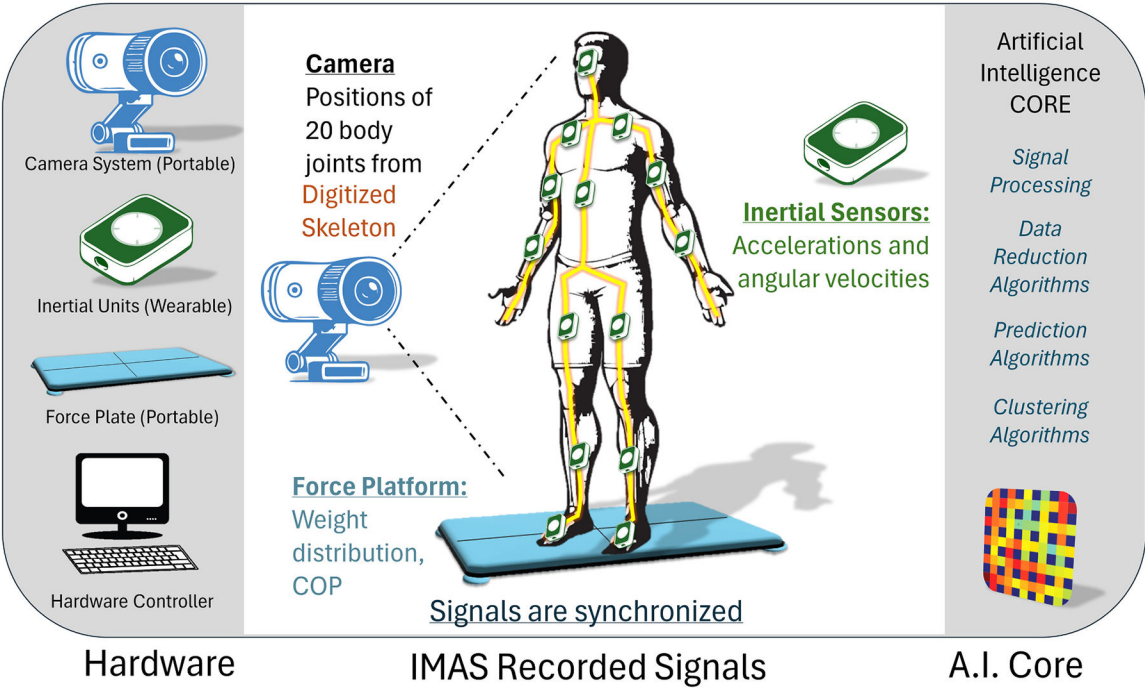


Fig. 1.
IMAS Hardware and Software. Modified from [28].

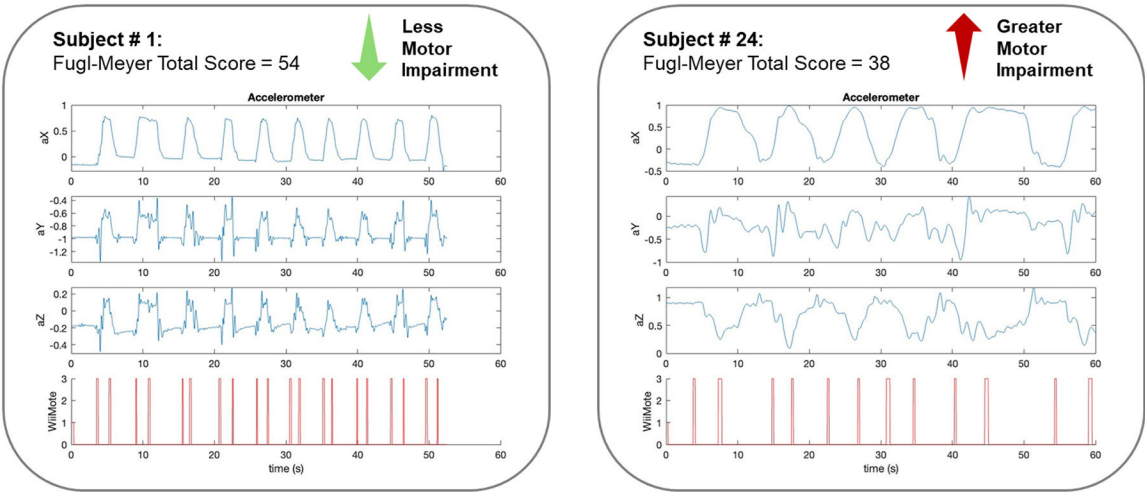


Fig. 2. Exemplary Data.
Data recorded from a wrist-mounted accelerometer during a task involving elbow flexion and extension. The recorded traces depict varying levels of motor abilities among the subjects.

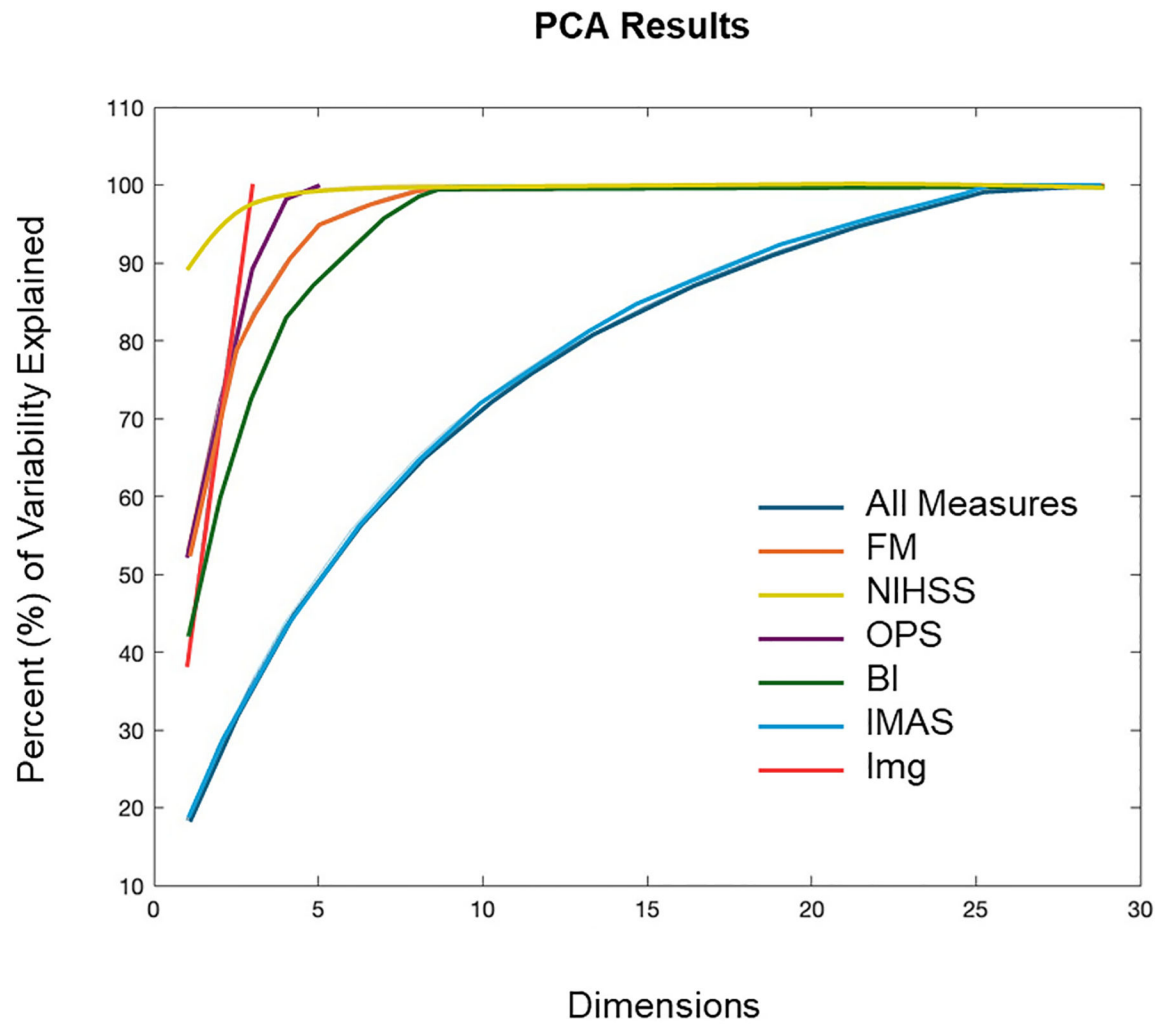


Fig. 3. PCA Results.

Figure abbreviations: FM = Fugl-Meyer, NIHSS=NIH Stroke Scale, OPS=Orpington Scale, BI = Barthel Index, IMAS = Integrated Motion Analysis Suite, Img = Imaging.

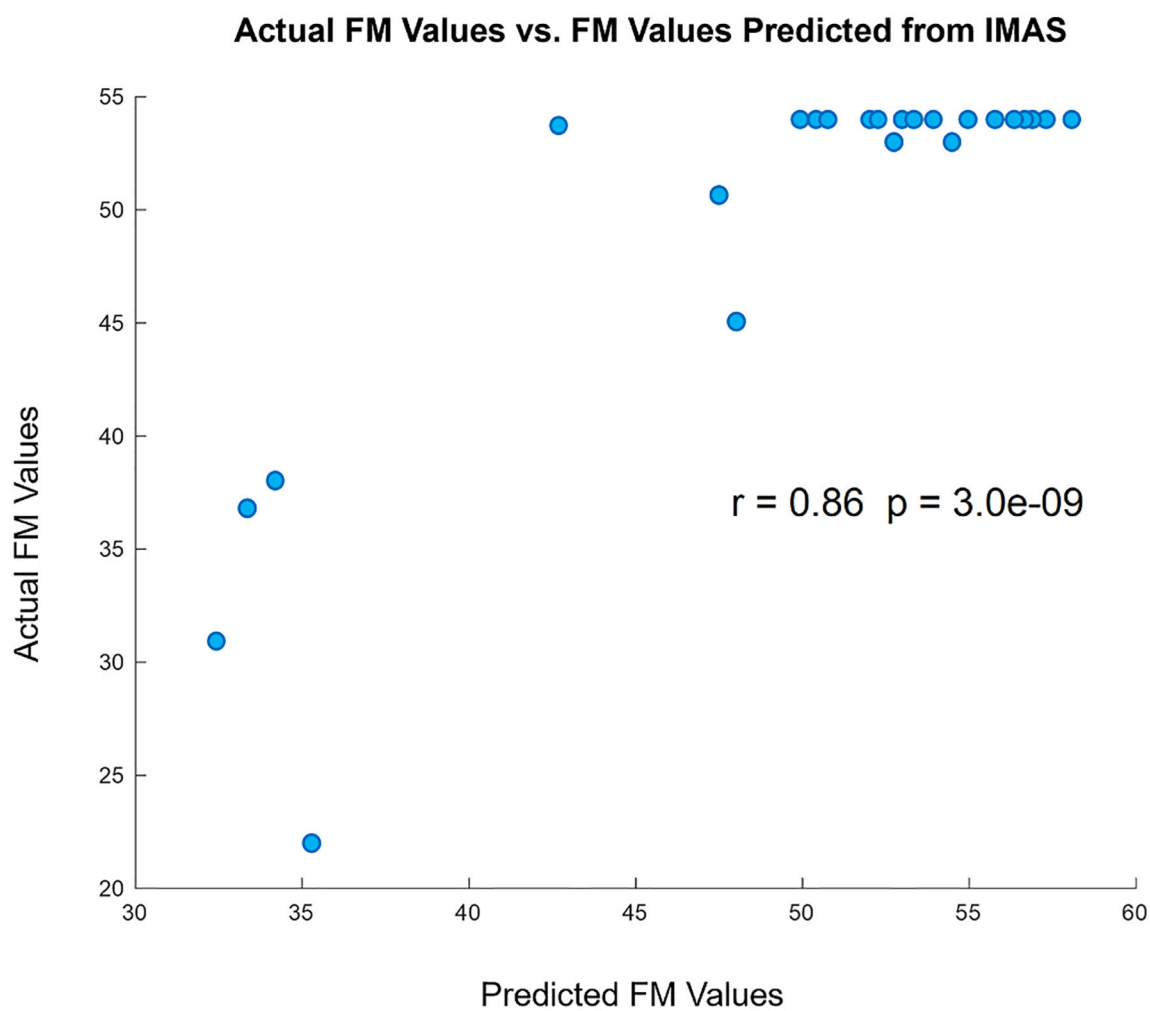


Fig. 4. FM Prediction Results.
Actual FM values vs. predicted FM values.

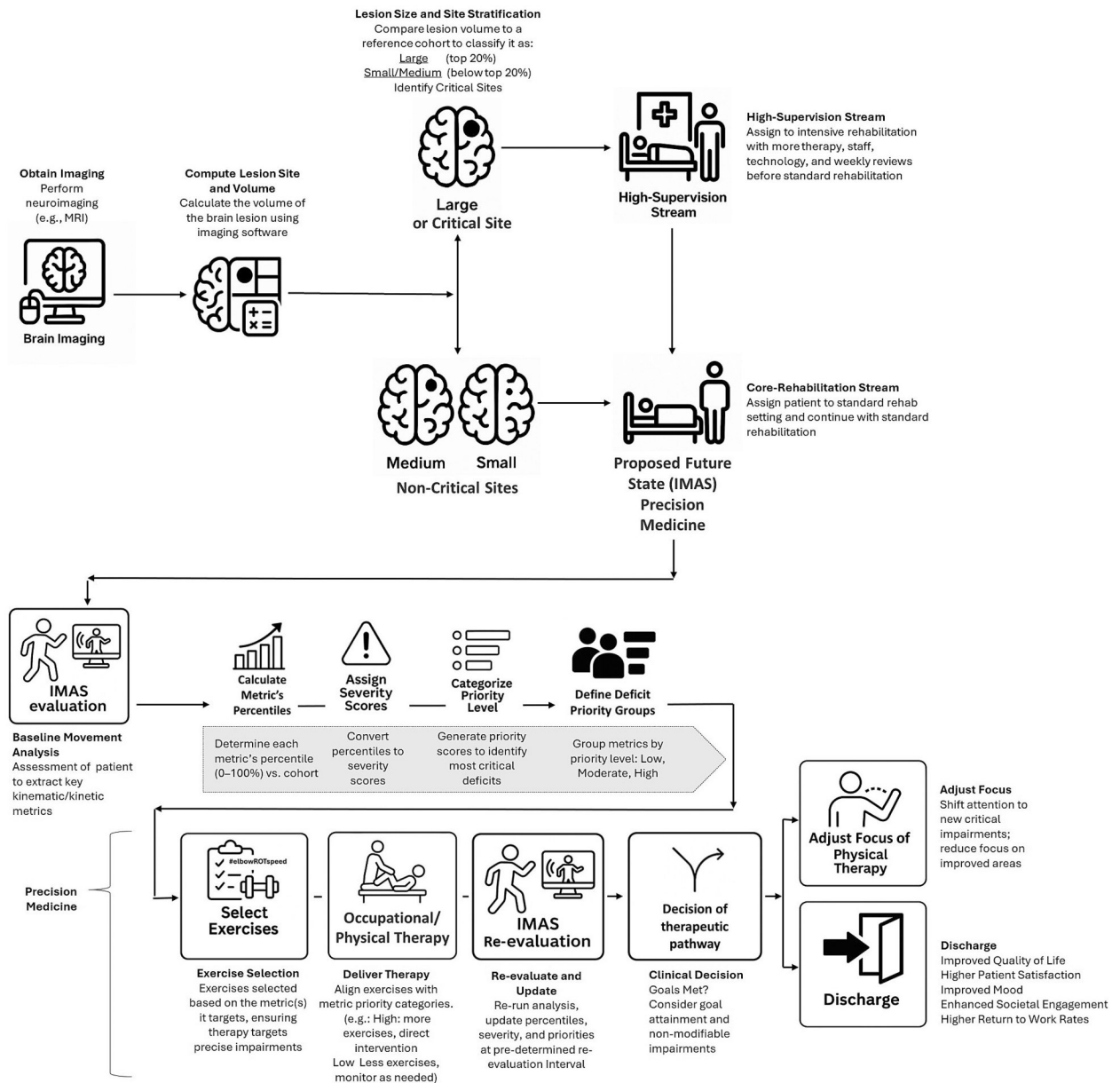


Fig. 5. Implementation of IMAS in the Clinical Workflow.

Implementation of the IMAS system within a clinical workflow to support motor recovery assessment and precision rehabilitation planning after stroke.

Table 1

Participants' characteristics.

Variable	Mean (SD) (or Count)
Age (years)	61.7 ± 12.2
Gender	21 males, 7 females
Race/Ethnicity	15 Black/African American (non-Hispanic)
	10 White (1 non-Hispanic, 9 Hispanic)
	1 American Indian/Alaskan Native (non-Hispanic)
	1 Hawaiian/Pacific Islander (non-Hispanic)
	1 Unknown ethnicity
Time from suspected stroke onset to hospital admission (hours)	20.2 ± 23.2 (range: 0.75 to 103.75)
Lesion Types	23 ischemic
	4 hemorrhagic
	1 combined
Average time to first evaluation (days)	5.6 ± 4.2
Upper Extremity FM Score	50 ± 8.1 (reflexes and coordination not assessed)

Table 2

Results of Predicting Models. The names for the predictors are as follows: EL = Elbow flexion/extension; SA=Shoulder abduction/adduction; SE=Shoulder flexion/extension; WA = Wrist abduction/adduction; WR = Wrist flexion/extension; FI = Finger flexion/extension; SY = FM assessment within synergies; HL = FM assessment Hand to lumbar; L = first part of the movement (e.g., flexion; see task description in Section 2.3); R = second part of the movement (e.g., extension; see task description in Section 2.3); Std = standard deviation; DS = path length; MeanRot = mean value of the angular speed; MaxRot = max value of the angular speed; MeanSpeed = mean value of speed; MaxSpeed = max value of speed; Duration = movement duration; Smoothness = movement smoothness; N = number of movements; meanAcc = mean acceleration magnitude; maxAcc = max value of acceleration magnitude. Descriptions of these tasks and metrics are provided in Section 2.3. β is the coefficient associated to each predictor, where a positive sign indicates a direct relationship with the outcome variable (e.g., higher values of the predictor correspond to better FM scores or recovery outcomes), while a negative sign indicates an inverse relationship (e.g., higher values of the predictor correspond to lower FM scores or poorer recovery outcomes).

Model Type	Predictors (β coefficient)	R ²	MAE	RMSE	RSR	VAF	Mean (SD)
FM from IMAS metrics only	IMAS_EL_RDS (-0.11)	0.75	2.82	4.03	0.496	74.42	50.01 (8.11)
	IMAS_EL_RStdDuration (-0.23)						
	IMAS_EL_RStdMaxRot (0.037)						
	IMAS_SA_N (0.08)						
	IMAS_SA_RStdMaxSpeed (-0.11)						
	IMAS_SA_RStdSmoothness (-0.10)						
	IMAS_SA_LStdMeanRot (-0.12)						
	IMAS_SE_RStdMeanRot (0.10)						
	IMAS_SY_LMeanRot (0.15)						
	IMAS_WA_LStdDuration (0.02)						
	IMAS_WR_LDuration (-0.18)						
	IMAS_WR_LStdMeanSpeed (0.02)						
	IMAS_WR_RStdMaxRot (0.08)						
	IMAS_FI_Number (0.21)						
OPS from IMAS metrics only	IMAS_SA_LMeanSpeed (-0.02)	0.58	0.46	0.55	0.642	57.31	2.85 (0.86)
	IMAS_SA_LDS (-0.24)						
	IMAS_SA_RSmoothness (-0.08)						
	IMAS_SE_RSmoothness (-0.06)						
	IMAS_SE_RDS (0.056)						
	IMAS_SE_LStdDuration (0.39)						
	IMAS_SY_LDS (0.03)						
	IMAS_SY_RDuration (0.03)						
	IMAS_WR_RMaxSpeed (-0.01)						
	IMAS_WR_RStdMaxRot (0.06)						
	IMAS_FI_Mean_R_dur_vect (0.002)						
	IMAS_FI_Std_R_mean_Rot (-0.02)						
	IMAS_FI_Std_R_mean_Acc (-0.09)						
	IMAS_FI_Std_R_max_Acc (-0.05)						

Model Type	Predictors (β coefficient)	R ²	MAE	RMSE	RSR	VAF	Mean (SD)
Barthel Index from IMAS metrics only	IMAS_EL_LDS (-0.28)	0.56	8.5	10.43	0.659	56.08	79.10 (15.81)
	IMAS_EL_LStdMeanRot (-0.04)						
	IMAS_EL_RStdMeanRot (-0.10)						
	IMAS_SA_LDS (0.09)						
	IMAS_SA_RStdSmoothness (-0.07)						
	IMAS_SA_RStdDS (-0.02)						
	IMAS_SA_RStdMeanRot (-0.002)						
	IMAS_SA_RStdMaxRot (-0.02)						
	IMAS_SE_LMeanSpeed (0.12)						
	IMAS_SE_LStdSmoothness (-0.008)						
	IMAS_SY_LDuration (-0.09)						
	IMAS_SY_RDuration (-0.12)						
	IMAS_WA_RMaxSpeed (0.15)						
	IMAS_FI_Mean_R_mean_Rot (0.13)						
Recovery from IMAS metrics only	IMAS_EL_RStdMaxSpeed (0.03)	0.83	2.60 %	0.03	0.416	82.46	96 % (7.7 %)
	IMAS_HL_MeanRot (0.06)						
	IMAS_SA_LStdMeanSpeed (-0.24)						
	IMAS_SA_RStdMaxSpeed (-0.16)						
	IMAS_SE_RDS (-0.36)						
	IMAS_SE_LStdSmoothness (0.07)						
	IMAS_SE_RStdMeanSpeed (0.05)						
	IMAS_SE_RStdMeanRot (0.10)						
	IMAS_SE_RStdMaxRot (-0.23)						
	IMAS_SY_LMeanRot (0.29)						
	IMAS_WR_RDS (0.10)						
	IMAS_WR_LMeanRot (-0.30)						
	IMAS_WR_RStdMeanRot (0.009)						
Recovery from IMAS and imaging metrics	IMAS_EL_RStdMaxSpeed (0.037)	0.89	2 %	0.02	0.327	88.97	96 % (7.7 %)
	IMAS_HL_MeanRot (0.06)						
	IMAS_SA_LStdMeanSpeed (-0.25)						
	IMAS_SA_RStdMaxSpeed (-0.14)						
	IMAS_SE_RDS (-0.33)						
	IMAS_SE_LStdSmoothness (0.08)						
	IMAS_SE_RStdMeanSpeed (0.03)						
	IMAS_SE_RStdMeanRot (0.16)						
	IMAS_SE_RStdMaxRot (-0.27)						
	IMAS_SY_LMeanRot (0.33)						
	IMAS_WR_RDS (0.10)						
	IMAS_WR_LMeanRot (-0.28)						
	IMAS_WR_RStdMeanRot (0.002)						
	IM_MRI_ischemic_vol (-0.07)						